AYMANE AIT DADS

Audio Machine Learning Intern | PyTorch · Mel Spectrograms · Transformer Fine-tuning

Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

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Hiring Team - Audio Machine Learning Intern
Bose Corporation · Framingham, MA, U.S.A

Dear Bose Hiring Team,

Bose has spent over 60 years proving that rigorous engineering and deep listening science are inseparable - and my work sits exactly at that intersection. At EURECOM, I built a **transformer autoencoder for unsupervised anomalous sound detection** on industrial audio, achieving **AUC 0.80** with zero labeled anomalies by learning directly from the structure of normal sound - the same principle that underlies acoustic modeling for intelligent audio products.

That project required hands-on signal processing decisions: I benchmarked **Mel spectrograms vs. MFCC** feature representations, applied **SpecAugment** data augmentation, and evaluated on a **DCASE-style protocol** matching state-of-the-art unsupervised baselines. Separately, I fine-tuned a **428M-parameter CLIP ViT-L/14** on 250K samples using **FP16 mixed precision and GradScaler**, running structured ablation experiments across unfreezing depth and learning rate schedules - ranking **#1 on the EURECOM private leaderboard**. Both projects share a discipline Bose demands: optimizing models under real hardware constraints without sacrificing generalization.

Bose's investment in psychoacoustics-driven ML - from adaptive noise cancellation in QuietComfort Ultra to research into personalized audio - makes this the one team where audio signal understanding and deep learning engineering are treated as a single problem, not two separate ones. I would welcome the chance to discuss how my background maps to your current research priorities.

Sincerely,

AYMANE AIT DADS